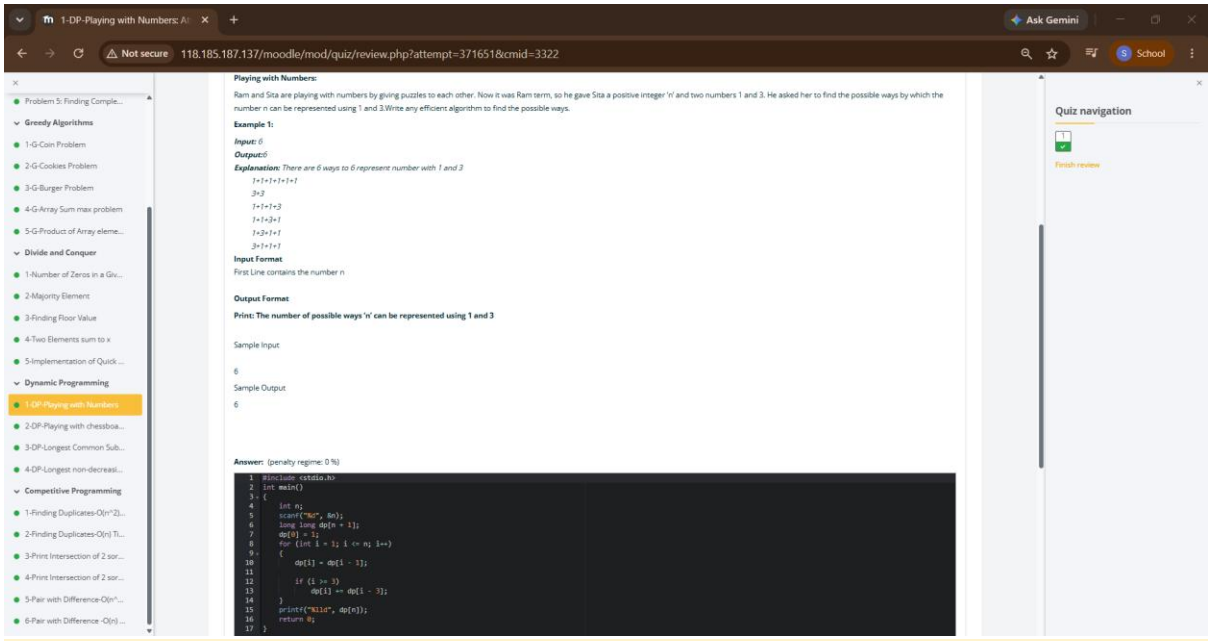


1-DP-Playing with Numbers: Ar

Not secure 118.185.187.137/moodle/mod/quiz/review.php?attempt=371651&cmid=3322

Ask Gemini

School

Problem 5: Finding Comple...
Greedy Algorithms
1-G-Coin Problem
2-G-Cookies Problem
3-G-Burger Problem
4-G-Array Sum max problem
5-G-Product of Array eleme...
Divide and Conquer
1-Number of Zeros in a Giv...
2-Majority Element
3-Finding Floor Value
4-Two Elements sum to x
5-Implementation of Quick ...
Dynamic Programming
1-DP-Playing with Numbers
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4-DP-Longest non-decreasi...
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1-Finding Duplicates-O(n^2)...
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3-Print Intersection of 2 sor...
4-Print Intersection of 2 sor...
5-Pair with Difference-O(n^...
6-Pair with Difference -O(n) ...

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    long long dp[n + 1];
    dp[0] = 1;
    for (int i = 1; i <= n; i++)
    {
        dp[i] = dp[i - 1];

        if (i >= 3)
            dp[i] += dp[i - 3];
    }
    printf("%lld", dp[n]);
    return 0;
}
```

Quiz navigation

1

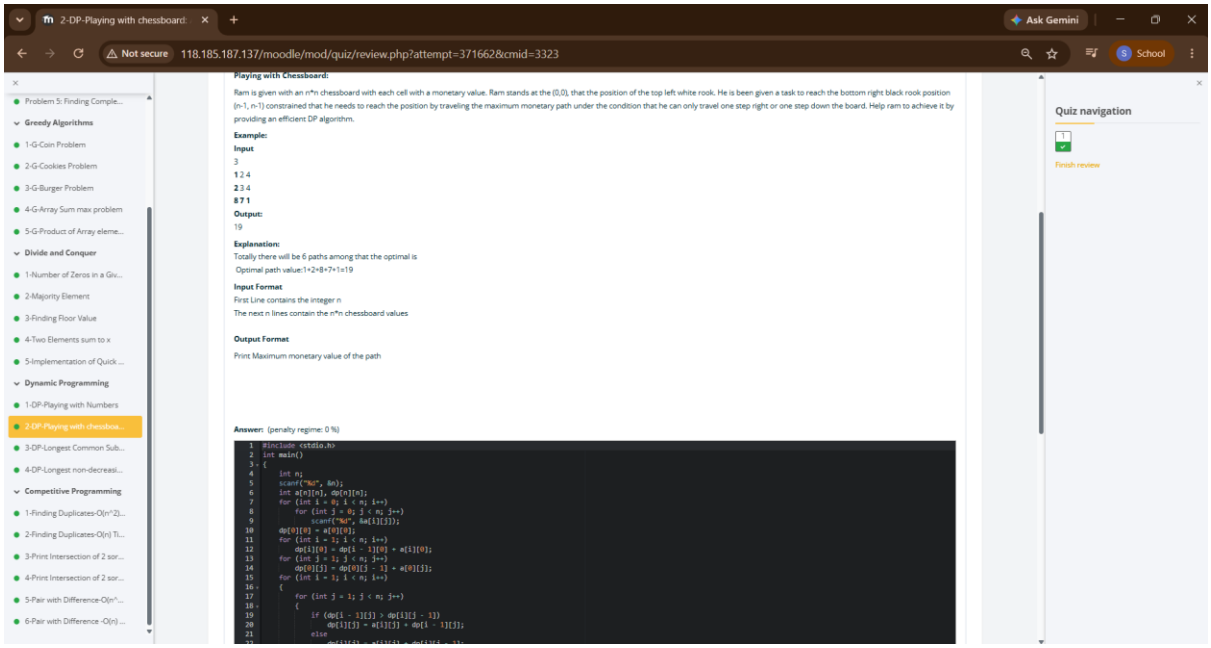
Finish review

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	2438281959672162 9	2438281959672162 9	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.



2-DP-Playing with chessboard:

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Problem 5: Finding Comple...
Greedy Algorithms
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Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:
Input
3
1 2 4
2 3 4
8 7 1
Output:
19

Explanation:
Totally there will be 6 paths among that the optimal is
Optimal path value:1+2+8+7+1=19

Input Format
First Line contains the integer n
The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

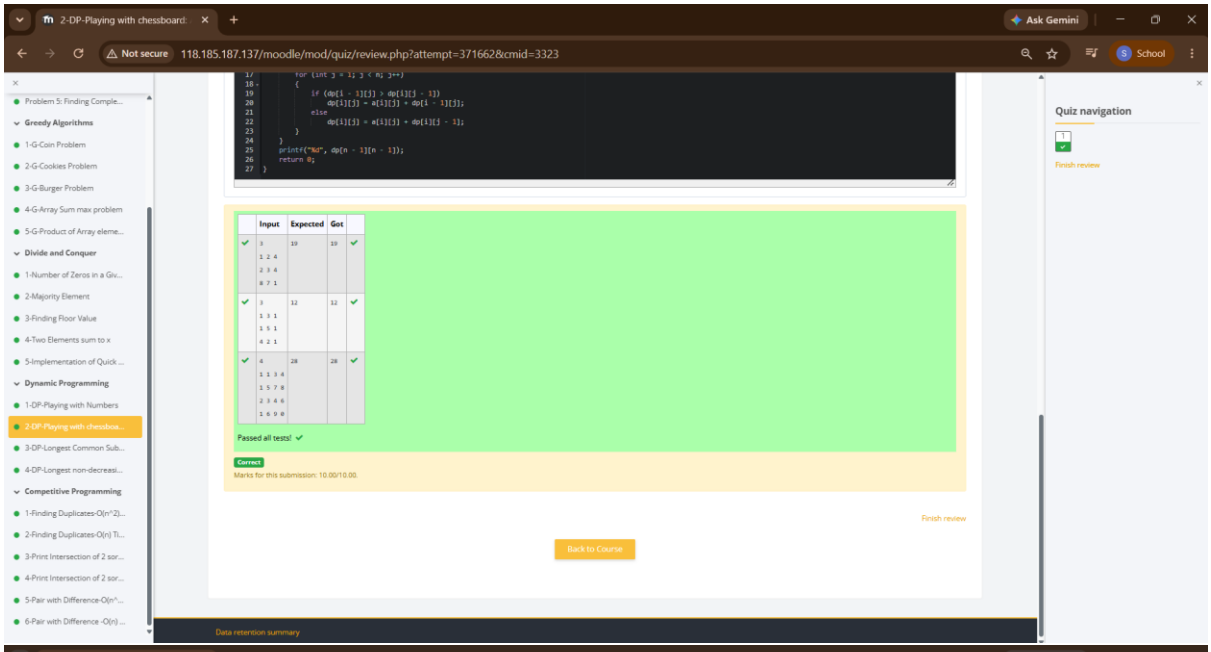
Answer: (penalty regime: 0 %)

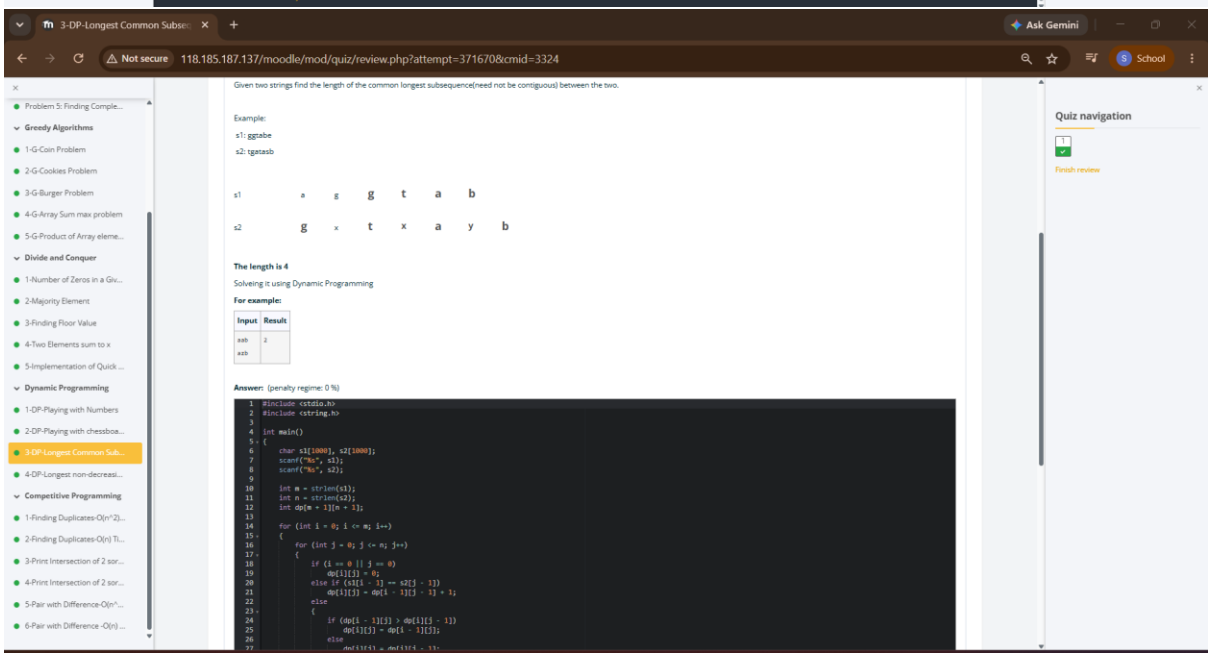
```c
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    int a[n][n], dp[n][n];
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            scanf("%d", &a[i][j]);
    dp[0][0] = a[0][0];
    for (int i = 1; i < n; i++)
        dp[i][0] = dp[i - 1][0] + a[i][0];
    for (int j = 1; j < n; j++)
        dp[0][j] = dp[0][j - 1] + a[0][j];
    for (int i = 1; i < n; i++)
    {
        for (int j = 1; j < n; j++)
        {
            if (dp[i - 1][j] > dp[i][j - 1])
                dp[i][j] = a[i][j] + dp[i - 1][j];
            else
```

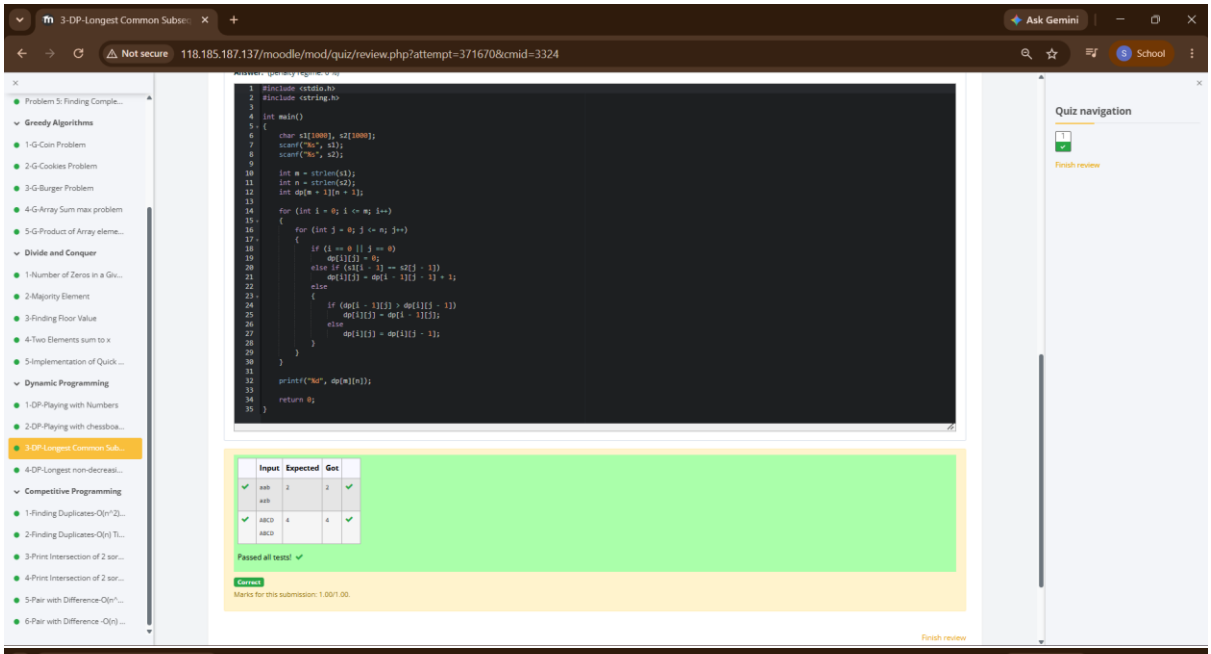
Quiz navigation

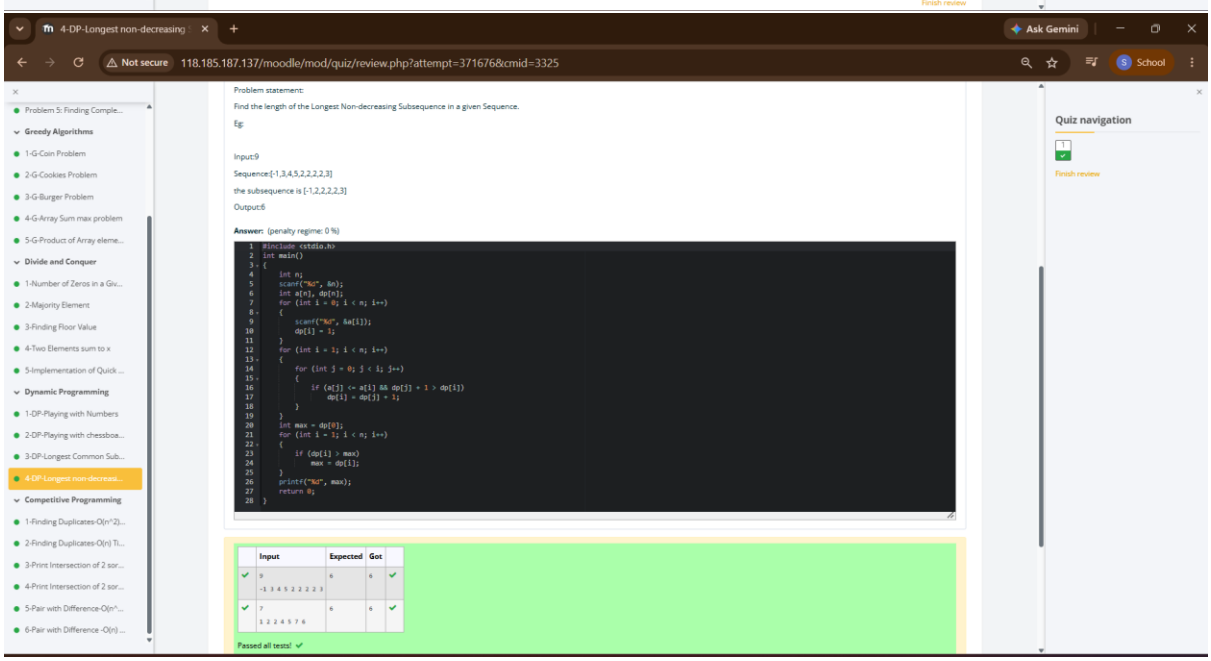
1

Finish review


2-DP-Playing with chessboard:
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=371662&cmid=3323
Ask Gemini
School
Problem 5: Finding Comple...
Greedy Algorithms
1-G-Coin Problem
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5-Pair with Difference-O(n^...
6-Pair with Difference -O(n) ...
for (int j = 1; j < m; j++)
{
if (dp[i - 1][j] > dp[i][j - 1])
dp[i][j] = a[i][j] + dp[i - 1][j];
else
dp[i][j] = a[i][j] + dp[i][j - 1];
}
}
printf("%d", dp[n - 1][m - 1]);
return 0;
}
Input
Expected
Got
3
1 2 4
2 3 4
8 7 1
19
19
3
1 3 1
1 5 1
4 2 1
12
12
4
1 1 3 4
1 5 7 8
2 3 4 6
1 6 9 8
28
28
Passed all tests!
Correct
Marks for this submission: 10.00/10.00.
Finish review
Back to Course
Data retention summary
Quiz navigation
Finish review


3-DP-Longest Common Subse:
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=371670&cmid=3324
Ask Gemini
School
Problem 5: Finding Comple...
Greedy Algorithms
1-G-Coin Problem
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2-Finding Duplicates-O(n) Ti...
3-Print Intersection of 2 sor...
4-Print Intersection of 2 sor...
5-Pair with Difference-O(n^...
6-Pair with Difference -O(n) ...
Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.
Example:
s1: ggtabe
s2: tgatasb
s1 a g g t a b
s2 g x t x a y b
The length is 4
Solveing it using Dynamic Programming
For example:
Input
Result
aab
azb
2
Answer: (penalty regime: 0 %)
#include <stdio.h>
#include <string.h>
int main()
{
char s1[1000], s2[1000];
scanf("%s", s1);
scanf("%s", s2);
int m = strlen(s1);
int n = strlen(s2);
int dp[m + 1][n + 1];
for (int i = 0; i <= m; i++)
{
for (int j = 0; j <= n; j++)
{
if (i == 0 || j == 0)
dp[i][j] = 0;
else if (s1[i - 1] == s2[j - 1])
dp[i][j] = dp[i - 1][j - 1] + 1;
else
{
if (dp[i - 1][j] > dp[i][j - 1])
dp[i][j] = dp[i - 1][j];
Quiz navigation
Finish review


3-DP-Longest Common Subsec
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=371670&cmid=3324
Ask Gemini
School
Problem 5: Finding Comple...
Greedy Algorithms
1-G-Coin Problem
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4-Print Intersection of 2 sor...
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6-Pair with Difference -O(n) ...
#include <stdio.h>
#include <string.h>
int main()
{
char s1[1000], s2[1000];
scanf("%s", s1);
scanf("%s", s2);
int m = strlen(s1);
int n = strlen(s2);
int dp[m + 1][n + 1];
for (int i = 0; i <= m; i++)
{
for (int j = 0; j <= n; j++)
{
if (i == 0 || j == 0)
dp[i][j] = 0;
else if (s1[i - 1] == s2[j - 1])
dp[i][j] = dp[i - 1][j - 1] + 1;
else
{
if (dp[i - 1][j] > dp[i][j - 1])
dp[i][j] = dp[i - 1][j];
else
dp[i][j] = dp[i][j - 1];
}
}
}
printf("%d", dp[m][n]);
return 0;
}
Input
Expected
Got
aab
azb
2
2
ABCD
AEBD
4
4
Passed all tests!
Correct
Marks for this submission: 1.00/1.00.
Quiz navigation
Finish review


4-DP-Longest non-decreasing
Not secure
118.185.187.137/moodle/mod/quiz/review.php?attempt=371676&cmid=3325
Ask Gemini
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Problem statement:
Find the length of the Longest Non-decreasing Subsequence in a given Sequence.
Eg:
Input:9
Sequence:[-1,3,4,5,2,2,2,2,3]
the subsequence is [-1,2,2,2,2,3]
Output:6
Answer: (penalty regime: 0 %)
#include <stdio.h>
int main()
{
int n;
scanf("%d", &n);
int a[n], dp[n];
for (int i = 0; i < n; i++)
{
scanf("%d", &a[i]);
dp[i] = 1;
}
for (int i = 1; i < n; i++)
{
for (int j = 0; j < i; j++)
{
if (a[j] <= a[i] && dp[j] + 1 > dp[i])
dp[i] = dp[j] + 1;
}
}
int max = dp[0];
for (int i = 1; i < n; i++)
{
if (dp[i] > max)
max = dp[i];
}
printf("%d", max);
return 0;
}
Input
Expected
Got
9
-1 3 4 5 2 2 2 2 3
6
6
7
1 2 2 4 5 7 6
6
6
Passed all tests!
Quiz navigation
Finish review